

Large–Prime Cofactor Descent for a Four–Möbius Walsh Energy

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July 2026

Abstract

The “four–Möbius gate” is the nonnegative equality energy left after expanding two source and two target Möbius factors in a fixed-shift prime-pair decomposition. Earlier papers in the twin-prime-correlation (TPC) sequence represented this energy as a Walsh polynomial but left its deterministic all–minus corner unresolved. We study that corner for the actual labels $u = d(\ell dj + h)$. If a prime in the global band $\sqrt{N_+} < p \leq N_-/R$ divides a target, then $\ell dj + h = pr$ with $r \geq R$, and setting its sign to -1 replaces $u = p(dr)$ by dr exactly. Equal descended labels have maximal multiplicity $\rho_R \leq X^{o(1)}(1 + L_0/R)$; the actual multiplicity, rather than this ambient upper bound, is the exact squared norm of the grouping map. Combining the descent with robust Walsh-face compression yields an unconditional band/cutoff tradeoff. In particular, the cutoff $(\log X)^{501/500}$ survives while the physical band reaches exponent $55021/105000 = 0.5240095\dots$, and the touched component alone admits $(\log X)^{1003/1000}$. At $y = X^{267/400+\eta}$, the complementary very-high-prime sector has an exact affine Gram form; low-sign averaging gives a second, pure Gram synthesis over equal residual labels. The remaining collisions lie on the fixed-determinant ladders $rp - dj\ell = h$ and have coherent physical Möbius sign. The closest dispersion and Kloosterman-fraction results reviewed here do not supply the resulting fixed- h , three-weight, vector-valued interface. No Möbius cancellation estimate, parity breach, prime-pair asymptotic, or twin-prime theorem is claimed.

Keywords: Möbius correlation; parity problem; large prime factor; cofactor descent; Walsh spectrum; Gram synthesis; determinant equation.

Convention. The shift $h \neq 0$ is fixed. All packets and prime sets at one scale are finite. Constants may depend on the fixed dyadic supports and on h , but not on X . We use $P^+(1) = 1$ and $P^-(1) = +\infty$.

MSC 2020: Primary 11N35, 11N36; Secondary 42C10, 11L07, 60E15.

1 Introduction

The terminal obstruction in the present TPC chain is no longer a formal sieve remainder. TPC–43 represented the surviving equality energy as

$$Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\kappa > 1} C_X(\kappa) \chi_\kappa(\varepsilon), \quad Z_0(-\mathbf{1}) = \mathcal{D}_0 + \sum_{\kappa > 1} \mu(\kappa) C_X(\kappa), \quad (1)$$

where the second value is the literal four–Möbius equality term, not a random-model surrogate [5]. TPC–44 proved that a density-one set of high-prime environments remains admissible after every sign

$$p \leq (\log X)^{501/500} \quad (2)$$

is overwritten arbitrarily, hence in particular by its physical value -1 [6]. It also proved that degree two, vanishing relative variance, and vanishing normalized influences do not select the prescribed all-minus corner. Thus a further abstract Boolean-cube argument cannot by itself cross the next gate.

This paper returns to the coefficients that generated (1). The equality labels are

$$u_{\alpha,j} = d_{\alpha}n_{\alpha,j} = d_{\alpha}(\ell_{\alpha}d_{\alpha}j + h), \quad (3)$$

with

$$\begin{aligned} Q &= X^{267/400+o(1)}, & J &= X^{133/400+o(1)}, \\ D_0 &= X^{10049/52500+o(1)}, & L_0 &:= Q/D_0 = X^{\lambda_L+o(1)}, & \lambda_L &:= \frac{99979}{210000}. \end{aligned} \quad (4)$$

We also write

$$\lambda_0 := 1 + \frac{10049}{52500} = \frac{62549}{52500} \quad (5)$$

for the exponent of the original fused labels. Here $m_{\alpha} = \ell_{\alpha}d_{\alpha} \asymp Q$, $\ell_{\alpha} \asymp L_0$, $d_{\alpha} \asymp D_0$, and $j \asymp J$. We retain every inherited mask, complex source weight, and output coordinate. No sign or equidistribution hypothesis is imposed on those deterministic coefficients.

1.1 The first positive result: a physical prime band

Let N_- and N_+ be the minimum and maximum positive equality targets $n_{\alpha,j}$, and let ℓ_+ be the largest active source prime. Define the packet-dependent band

$$\mathcal{P}_{\text{bal}} := \left\{ p \text{ prime} : \sqrt{N_+} < p \leq \frac{N_-}{\ell_+} \right\}. \quad (6)$$

At the endpoint it occupies, up to fixed support margins, the exponent window

$$\frac{1}{2} < \frac{\log p}{\log X} < 1 - \frac{99979}{210000} = \frac{110021}{210000}; \quad (7)$$

its width is

$$\frac{5021}{210000} = 0.0239095\dots \quad (8)$$

This is a genuinely growing prime interval, not a finite set of coordinates.

If $p \in \mathcal{P}_{\text{bal}}$ divides an active target, write

$$n_{\alpha,j} = pr, \quad s = d_{\alpha}r. \quad (9)$$

The prime p is unique, $r \geq \ell_+$, and

$$\mu(u_{\alpha,j}) = \mu(ps) = -\mu(s). \quad (10)$$

Thus setting every sign in $\mathcal{P}_{\text{bal}}$ to its physical value deletes that prime exactly from the character. Equal residual labels have at most

$$O_h(\tau(s)) = X^{o(1)} \quad (11)$$

preimages in a fixed Hilbert coordinate. The proof is elementary but uses the actual equation

$$rp - d_{\alpha}j\ell_{\alpha} = h. \quad (12)$$

After grouping collisions, the new diagonal is at most $X^{o(1)}\mathcal{D}_0$.

Combining this descent with the robust-face theorem of TPC-44 yields our principal de-randomization statement. For a density $1 - o(1)$ set of the remaining prime-sign environments, all signs in the entire band $\mathcal{P}_{\text{bal}}$ may be fixed simultaneously to -1 , while all signs in (2) may still be overwritten arbitrarily, and the resulting equality energy remains within the inherited $K_B = X^{1/400+o(1)}$ budget. This is the first step in this chain that physically freezes a growing large-prime band. More generally, if $R = X^{\theta+o(1)}$ and $z = (\log X)^A$, the sufficient frontier is

$$(\lambda_L - \theta)_+ + \lambda_0(1 - 1/A) < \frac{1}{400}. \quad (13)$$

For example, retaining $A = 501/500$ and taking $R = \ell_+ X^{-1/10000}$ extends the upper band exponent to

$$1 - \lambda_L + \frac{1}{10000} = \frac{55021}{105000} = 0.5240095\dots, \quad (14)$$

with exact unused margin $577/26302500$. These statements do not freeze the intervening or still larger primes.

1.2 Two complementary large-prime normal forms

The band theorem is most effective when the cofactor is at least the source-prime length. Two further exact statements locate what happens on either side of that balance.

First, the touched residual labels satisfy

$$s \leq D_0 \sqrt{N_+} = X^{\lambda_1+o(1)}, \quad \lambda_1 = \frac{1}{2} + \frac{10049}{52500} = \frac{36299}{52500}. \quad (15)$$

For the touched component alone, TPC-44's Rankin projection argument therefore admits every

$$1 < A < A_1^* := \left(1 - \frac{1}{400\lambda_1}\right)^{-1} = \frac{145196}{144671}. \quad (16)$$

We choose $A = 1003/1000$. The exact endpoint margin is

$$\frac{1}{400} - \lambda_1 \left(1 - \frac{1000}{1003}\right) = \frac{30329}{70210000} > 0. \quad (17)$$

This improves the robust low-prime cutoff of the shortened component to $(\log X)^{1003/1000}$. Untouched labels retain their original size, so this larger cutoff is not asserted for the full packet.

Second, fix $\eta > 0$ and move to

$$y = X^{267/400+\eta}. \quad (18)$$

Every equality label then contains either no prime above y or one such prime, and fixed (j, p) meets at most one source row in each source system. With all lower signs placed at their physical values, the complete equality packet becomes

$$Z_0(-\mathbf{1}_{\leq y}, \varepsilon_{> y}) = \left\| v_0 + \sum_{p>y} \varepsilon_p v_p \right\|^2. \quad (19)$$

Thus the large-prime Walsh hypergraph is exactly a Gram graph: its degree is at most two, and the general coefficient descent of TPC-44 reduces to correlations among the vectors v_p .

Averaging, rather than fixing, the signs below y gives an even sharper rough projection. For explicitly lifted residual vectors x_p ,

$$\mathbb{E}_{\leq y} Z_0(\varepsilon_{\leq y}, -\mathbf{1}_{> y}) = \mathcal{D}_0 - \mathcal{D}_H + \left\| \sum_{p>y} x_p \right\|^2, \quad \mathcal{D}_H = \sum_{p>y} \|x_p\|^2 \leq \mathcal{D}_0. \quad (20)$$

The remaining off-diagonal collisions are exactly equal deleted-label fibers. Writing $n = pr$, each atom on one such fiber lies on (12). When $r \ll L_0$, the ambient two-prime ladder has length $O_h(1 + L_0/r)$, and all atoms with the same residual label have the same Möbius sign. On every realized fiber, Cauchy–Schwarz is sharp for arbitrary deterministic coefficients; the ambient ladder bound need not be attained.

1.3 What is and is not proved

The unconditional advances are structural and de-randomizing:

- (i) the actual coefficients are expanded without an auxiliary sign assumption;
- (ii) a full mesoscopic band of prime signs is fixed physically at subpolynomial diagonal cost;
- (iii) the shortened component receives a stronger low-prime cutoff;
- (iv) the very-large-prime part becomes an exact affine Gram graph;
- (v) the remaining small-cofactor obstruction is identified as a fixed-determinant, coefficient-specific synthesis estimate.

We do *not* prove the synthesis estimate in (20); we do not deduce the complete physical equality bound; and we do not obtain a prime-pair asymptotic, a parity breakthrough, or a bounded-gap statement. The literature audit in section 10 explains a particularly precise reason: the closest cited fixed-determinant theorem permits two arbitrary arithmetic coefficient sequences, whereas the TPC packet has three nonsmooth arithmetic slots after the descent.

Sections 2 and 3 reconstruct the actual coefficient and incidence geometry. The physical-band descent and its sharp fiber estimate are proved in sections 4 and 5. Section 6 gives the two robust freezing consequences. The affine and rough Gram forms are derived in section 7; the fixed-determinant ladders and exact support boundary appear in sections 8 and 9. We compare the result with current analytic inputs in section 10, state the next gate in section 11, and close with a machine-checkable algebraic certificate and claim ledger.

2 The actual equality coefficient interface

This section removes an ambiguity that is harmless in an abstract Walsh argument but not at the present coefficient-specific gate. The Walsh amplitudes are exactly the TPC terminal atoms after the two Möbius values in each row have been stripped; they are not new free coefficients.

2.1 Rows, outputs, and stripped atoms

Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be the two inherited source-row systems. A row $\rho \in \mathcal{A}_s$ is

$$\rho = (\ell_\rho, d_\rho), \quad m_\rho = \ell_\rho d_\rho \asymp Q, \quad \ell_\rho \asymp L_0, \quad d_\rho \asymp D_0, \quad (21)$$

where ℓ_ρ is prime, d_ρ is squarefree, $(d_\rho, h) = 1$, and m_ρ determines ρ . Let $\mathcal{J} \subset \mathbb{N}$ be the finite orbit-time set, supported on $j \asymp J$. Here Φ_0 is the inherited terminal cutoff, $\gamma_\rho^{\circ, (s)} \in \mathbb{C}$ is the stripped source weight, and $\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) \in \mathbb{C}$ is the joint mask; only their support and the size bounds below are used [5, 6]. Put

$$n_{\rho, j} := m_\rho j + h = \ell_\rho d_\rho j + h, \quad \Lambda(n) := (\log n) \Phi_0(n). \quad (22)$$

All atoms below are zero outside the inherited positive terminal support. On that support

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad D_0 = X^{10049/52500+o(1)}, \quad QJ \asymp X. \quad (23)$$

The source signs factor as

$$\gamma_\rho^{(s)} = \mu(d_\rho) \gamma_\rho^{\circ, (s)}, \quad s = 1, 2. \quad (24)$$

The left output coordinates and right output coordinates are, respectively,

$$\mathcal{I}_L := \{(L, \gamma, j) : \gamma \in \mathcal{A}_2, j \in \mathcal{J}\}, \quad \mathcal{I}_R := \{(R, \alpha, j) : \alpha \in \mathcal{A}_1, j \in \mathcal{J}\}, \quad (25)$$

and we write

$$\mathcal{H}_{\text{out}} := \ell^2(\mathcal{I}_L \sqcup \mathcal{I}_R). \quad (26)$$

Thus $\alpha \in \mathcal{A}_1$ is the summation row in a left coordinate, whereas $\gamma \in \mathcal{A}_2$ is the summation row in a right coordinate. Define the actual stripped atoms

$$B_{\alpha; \gamma, j}^L := -\gamma_\alpha^{\circ, (1)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) \Lambda(n_{\alpha, j}) \mu^2(n_{\alpha, j}), \quad (27)$$

$$B_{\gamma; \alpha, j}^R := -\gamma_\gamma^{\circ, (2)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) \Lambda(n_{\gamma, j}) \mu^2(n_{\gamma, j}). \quad (28)$$

The inherited coefficient and support bounds used below are

$$\max_{\alpha, \gamma, j} \{|B_{\alpha; \gamma, j}^L|, |B_{\gamma; \alpha, j}^R|\} \leq X^{o(1)}, \quad \#\mathcal{A}_s \leq X^{o(1)} Q \quad (s = 1, 2). \quad (29)$$

The semicolon records the summation row on its left and the opposite output row on its right. In particular, it prevents the right-hand indexing from being mistaken for a second copy of the left one.

For every nonzero atom, $n_{\rho, j}$ is squarefree and

$$(d_\rho, n_{\rho, j}) = (d_\rho, h) = 1. \quad (30)$$

Its fused equality label is therefore the squarefree integer

$$\boxed{u_{\rho, j} := d_\rho n_{\rho, j} = d_\rho(\ell_\rho d_\rho j + h)}. \quad (31)$$

Lemma 2.1 (Fixed-coordinate label injectivity). *For all sufficiently large X , fixed j , and physical rows ρ, σ in the same source system,*

$$u_{\rho, j} = u_{\sigma, j} \implies \rho = \sigma. \quad (32)$$

Proof. Equality gives

$$j(\ell_\rho d_\rho^2 - \ell_\sigma d_\sigma^2) = h(d_\sigma - d_\rho). \quad (33)$$

If the integer in parentheses is nonzero, the left side has absolute value at least j , while the right side is at most $|h| \max d_\tau < j$, because $D_0 = o(J)$. Hence both sides vanish. Since $h \neq 0$, first $d_\rho = d_\sigma$, then $\ell_\rho = \ell_\sigma$. $\square$

Let $\chi_u(\varepsilon) = \prod_{p|u} \varepsilon_p$. The exact left and right coordinate polynomials are

$$S_{L, \gamma, j}(\varepsilon) = \sum_{\alpha \in \mathcal{A}_1} B_{\alpha; \gamma, j}^L \chi_{u_{\alpha, j}}(\varepsilon), \quad (34)$$

$$S_{R, \alpha, j}(\varepsilon) = \sum_{\gamma \in \mathcal{A}_2} B_{\gamma; \alpha, j}^R \chi_{u_{\gamma, j}}(\varepsilon). \quad (35)$$

Equivalently, the abstract coefficient $b_{i, u}$ is

$$b_{(L, \gamma, j), u_{\alpha, j}} = B_{\alpha; \gamma, j}^L, \quad b_{(R, \alpha, j), u_{\gamma, j}} = B_{\gamma; \alpha, j}^R, \quad (36)$$

and is zero for absent labels. The injectivity lemma makes (36) single-valued. Define the active-label set and the high part by

$$\mathcal{U}_i := \{u : b_{i,u} \neq 0\}, \quad u_{>y} := \prod_{\substack{p|u \\ p>y}} p \quad \text{for every cutoff } y. \quad (37)$$

With this notation, the equality energy and atomic diagonal are

$$\begin{aligned} S(\varepsilon) &:= (S_i(\varepsilon))_{i \in \mathcal{I}_L \sqcup \mathcal{I}_R} \in \mathcal{H}_{\text{out}}, \\ Z_0(\varepsilon) &:= \|S(\varepsilon)\|^2 = \sum_{i \in \mathcal{I}_L \sqcup \mathcal{I}_R} |S_i(\varepsilon)|^2, \\ \mathcal{D}_0 &:= \sum_i \sum_u |b_{i,u}|^2. \end{aligned} \quad (38)$$

At the all-minus point, $\chi_{u_{\rho,j}}(-\mathbf{1}) = \mu(d_\rho)\mu(n_{\rho,j})$, so (34) and (35) recover the literal physical columns. No Möbius sign remains hidden in $B^{L/R}$.

2.2 The actual Walsh coefficient

For every positive integer define its parity kernel by

$$\text{sf}(N) := \prod_{\substack{p \\ v_p(N) \text{ odd}}} p. \quad (39)$$

In particular, if a, b are squarefree, then

$$\text{sf}(ab) = \frac{ab}{(a, b)^2}. \quad (40)$$

For two rows at the same orbit time put

$$\kappa_{\rho, \sigma, j} := \text{sf}(u_{\rho, j} u_{\sigma, j}). \quad (41)$$

Proposition 2.2 (Complete original-variable formula for C_X). *For $\kappa > 1$, the scale-dependent Walsh coefficient is*

$$C_X(\kappa) = C_X^L(\kappa) + C_X^R(\kappa), \quad (42)$$

where

$$C_X^L(\kappa) := \sum_{\gamma \in \mathcal{A}_2} \sum_{j \in \mathcal{J}} \sum_{\substack{\alpha, \beta \in \mathcal{A}_1, \alpha \neq \beta \\ \kappa_{\alpha, \beta, j} = \kappa}} B_{\alpha; \gamma, j}^L \overline{B_{\beta; \gamma, j}^L}, \quad (43)$$

$$C_X^R(\kappa) := \sum_{\alpha \in \mathcal{A}_1} \sum_{j \in \mathcal{J}} \sum_{\substack{\gamma, \delta \in \mathcal{A}_2, \gamma \neq \delta \\ \kappa_{\gamma, \delta, j} = \kappa}} B_{\gamma; \alpha, j}^R \overline{B_{\delta; \alpha, j}^R}. \quad (44)$$

Both inner sums are ordered. In particular,

$$C_X(\kappa) \in \mathbb{R}, \quad Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\kappa > 1} C_X(\kappa) \chi_\kappa(\varepsilon). \quad (45)$$

Proof. Expand every squared coordinate in (38). The diagonal row pairs give $\mathcal{D}_0$. By lemma 2.1, no unequal pair has constant character. An unequal ordered pair contributes $B_\rho \overline{B_\sigma} \chi_{\kappa_{\rho, \sigma, j}}$, which gives equations (43) and (44). Reversing the ordered pair conjugates its weight without changing its kernel, proving reality. $\square$

For example, a left summand in (43) equals

$$\gamma_{\alpha}^{\circ,(1)} \overline{\gamma_{\beta}^{\circ,(1)}} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) \overline{\mathfrak{A}_{\beta,\gamma}^{\text{act}}(j)} \Lambda(n_{\alpha,j}) \overline{\Lambda(n_{\beta,j})} \times \mu^2(n_{\alpha,j}) \mu^2(n_{\beta,j}). \quad (46)$$

The two leading minus signs in (27) have cancelled. There is no algebraic sign for (46); the reality of C_X is a conjugation symmetry, not positivity.

2.3 Restriction matching in original variables

Let $p_1 < \dots < p_R$ be the active primes of the centered Walsh spectrum, $P_r = \prod_{t \leq r} p_t$, and extend C_X by $C_X(1) = 0$. As in TPC-44, put

$$C_r(m) := \sum_{a|P_r} \mu(a) C_X(am), \quad (m, P_r) = 1. \quad (47)$$

To expand its correlation, let $\mathcal{E}_X$ be the disjoint union of all ordered unequal row pairs in equations (43) and (44). For $e \in \mathcal{E}_X$, write

$$w(e) := b_{i,u_{e,1}} \overline{b_{i,u_{e,2}}}, \quad \kappa(e) := \text{sf}(u_{e,1} u_{e,2}). \quad (48)$$

The two labels inside one e share its side, output coordinate, and orbit time. Two different atoms e, f , however, need not share any of those data.

Proposition 2.3 (Four-label matching formula). *For $p = p_r$, the TPC-44 paired coefficient sum has the exact form*

$$\begin{aligned} \mathcal{T}_r &:= \sum_{(m, P_r)=1} C_{r-1}(m) \overline{C_{r-1}(pm)} \\ &= \sum_{c|P_{r-1}} \mu(c) \sum_{\substack{e, f \in \mathcal{E}_X \\ p \nmid \kappa(e), p \nmid \kappa(f) \\ \text{sf}(u_{e,1} u_{e,2} u_{f,1} u_{f,2}) = pc}} w(e) \overline{w(f)}. \end{aligned} \quad (49)$$

Equivalently, every summand lies on an exact almost-square equation

$$u_{e,1} u_{e,2} u_{f,1} u_{f,2} = z^2 pc, \quad c \mid P_{r-1}, \quad (50)$$

for an integer z . In row variables, the left side is a product of four factors of the form

$$d_{\rho}(\ell_{\rho} d_{\rho} j + h), \quad (51)$$

with one common orbit time inside each of the two ordered pairs, but not necessarily between the pairs.

Proof. Insert (47) twice:

$$\mathcal{T}_r = \sum_m \sum_{a,b|P_{r-1}} \mu(a) \mu(b) \sum_{\kappa(e)=am} \sum_{\kappa(f)=bpm} w(e) \overline{w(f)}. \quad (52)$$

Here a, b, p, m have the coprimality conditions forced by their displayed factorizations. Put $c = \text{sf}(ab) = ab/(a, b)^2$. Then $c \mid P_{r-1}$,

$$\mu(a) \mu(b) = \mu(c), \quad \text{sf}((am)(bpm)) = pc. \quad (53)$$

Conversely, the oriented conditions in (49) say that the two kernels have the same part outside P_r , while precisely the second contains p . They therefore recover unique a, b, m in (52). Finally, squarefree reduction is addition of prime valuations modulo two, so

$$\text{sf}(\kappa(e) \kappa(f)) = \text{sf}(u_{e,1} u_{e,2} u_{f,1} u_{f,2}). \quad (54)$$

This proves both formulas. $\square$

The sign $\mu(c)$ in (49) is supplied only by the previously frozen primes. Neither $w(e)$ nor $w(f)$ has a prescribed sign. Consequently, an estimate for $\mathcal{T}_r$ requires cancellation or an angle in this actual weighted four-label matching problem; label incidence alone does not determine its sign.

3 Prime incidence in the actual fused labels

We next separate source-divisor incidence from target-affine incidence. This distinction is invisible in a generic squarefree label model and is the first arithmetic gain obtained by returning to (31).

Write the fixed dyadic supports as

$$D_- \leq d_\rho \leq D_+, \quad M_- \leq m_\rho \leq M_+, \quad J_- \leq j \leq J_+, \quad N_- \leq n_{\rho,j} \leq N_+, \quad (55)$$

where

$$D_\pm \asymp D_0, \quad M_\pm \asymp Q, \quad J_\pm \asymp J, \quad N_\pm \asymp X. \quad (56)$$

Put $\Delta_M = M_+ - M_- \ll Q$.

Lemma 3.1 (Exact source/target alternative). *Let $p > |h|$ be prime. For a nonzero equality atom, $p \mid u_{\rho,j}$ holds precisely in one of the following disjoint ways:*

(i) $p \mid d_\rho$, in which case $n_{\rho,j} \equiv h \pmod{p}$ and $p \nmid n_{\rho,j}$;

(ii) $p \nmid d_\rho m_\rho$ and

$$j \equiv -hm_\rho^{-1} \pmod{p}, \quad (57)$$

equivalently $p \mid n_{\rho,j}$.

In particular, the source prime ℓ_ρ does not divide its own label once $\ell_\rho > |h|$.

Proof. The label is $d_\rho n_{\rho,j}$, and the two factors are coprime by (30). If $p \mid d_\rho$, then $n_{\rho,j} = \ell_\rho d_\rho j + h \equiv h \not\equiv 0 \pmod{p}$. Otherwise incidence means $p \mid n_{\rho,j}$. A prime divisor of m_ρ would again give $n_{\rho,j} \equiv h \pmod{p}$, so m_ρ is invertible and (57) follows. Taking $p = \ell_\rho$ proves the last assertion. $\square$

Proposition 3.2 (One-dimensional incidence fibers). *Let $p > \max\{|h|, D_+\}$. Then*

$$\#\{\rho : p \mid u_{\rho,j}\} \leq 1 + \frac{\Delta_M}{p} \ll 1 + \frac{Q}{p} \quad \text{for fixed } j, \quad (58)$$

$$\#\{j \in \mathcal{J} : p \mid u_{\rho,j}\} \ll 1 + \frac{J}{p} \quad \text{for fixed } \rho. \quad (59)$$

Moreover,

$$\#\{(\rho, j) : p \mid u_{\rho,j}\} \ll X^{o(1)} \left(1 + \frac{X}{p}\right). \quad (60)$$

All statements hold separately in each source-row system.

Proof. Only the target alternative of lemma 3.1 remains. If two rows occur at fixed j , then $p \nmid j$, since otherwise $n_{\rho,j} \equiv h \pmod{p}$. Hence all their distinct integers m_ρ occupy the one class $-hj^{-1} \pmod{p}$ inside an interval of length Δ_M , proving (58). For a fixed row, (57) places j in one class in an interval of length $O(J)$, proving (59).

For the last bound, write

$$n_{\rho,j} = pr, \quad m_\rho j = pr - h. \quad (61)$$

There are $O(1 + X/p)$ possible positive cofactors r . For fixed (p, r) , every possible m_ρ divides $pr - h$; the orbit time is then $(pr - h)/m_\rho$, and source separation recovers at most one row from m_ρ . The uniform divisor bound $\tau(pr - h) = X^{o(1)}$ proves (60). $\square$

The proposition gives the following genuine three-stage geometry beyond the logarithmic cutoff of TPC-44:

- (a) for $p \lesssim D_0$, a source divisor $p \mid d_\rho$ can persist through every orbit time of a row;
- (b) for $D_0 \ll p \lesssim Q$, incidence is purely affine, with fixed-time multiplicity $O(1 + Q/p)$; and
- (c) for $p \gg Q$, fixed (j, p) meets at most one row.

Also, once $p \gg J$, a fixed row meets at most one orbit time.

3.1 The very-high cutoff

For a completely constant-free formulation, let $y = y_X$ satisfy

$$y > \max\{|h|, D_+, J_+, \Delta_M, \sqrt{N_+}\} \quad (62)$$

and

$$\frac{D_+ N_+}{y} < D_- N_- . \quad (63)$$

At the optimized endpoint, every fixed

$$0 < \eta < \frac{133}{400}, \quad \boxed{y = X^{267/400 + \eta}} \quad (64)$$

satisfies (62) and (63) for all sufficiently large X .

Theorem 3.3 (Unique very-high prime and singleton incidence). *Assume (62). Every nonzero equality label has the unique factorization*

$$u_{\rho,j} = s_{\rho,j} q_{\rho,j}, \quad q_{\rho,j} = 1 \text{ or a prime greater than } y, \quad P^+(s_{\rho,j}) \leq y. \quad (65)$$

If $q_{\rho,j} = p > y$, then

$$n_{\rho,j} = p r_{\rho,j}, \quad \boxed{s_{\rho,j} = d_\rho r_{\rho,j}}, \quad r_{\rho,j} \leq \frac{N_+}{p} \leq \frac{N_+}{y}. \quad (66)$$

For fixed (j, p) , at most one row in either given source system has $q_{\rho,j} = p$.

Proof. Every prime divisor of d_ρ is at most $D_+ < y$. Since $n_{\rho,j} \leq N_+ < y^2$, it has at most one prime factor above y . This proves (65). If that prime is p , squarefreeness and (30) give

$$u_{\rho,j} = d_\rho(p r_{\rho,j}) = p(d_\rho r_{\rho,j}), \quad (p, d_\rho r_{\rho,j}) = 1, \quad (67)$$

which proves the exact residual label in (66). In particular, it is $d_\rho r$, not $\ell_\rho d_\rho r$; the source prime occurs in the slope, not as a factor of the fused label.

Finally, if two rows at fixed j have targets divisible by p , then $p \nmid j$ and

$$p \mid (m_\rho - m_\sigma)j. \quad (68)$$

Thus $p \mid m_\rho - m_\sigma$. But $|m_\rho - m_\sigma| \leq \Delta_M < y < p$, so the two slopes, and hence the two source rows, coincide. $\square$

Corollary 3.4 (Very-high atomic mass). *Let*

$$\mathcal{D}_p := \sum_i \sum_{\substack{u \in \mathcal{U}_i \\ u > y = p}} |b_{i,u}|^2. \quad (69)$$

Then

$$\sum_{p>y} \mathcal{D}_p \leq \mathcal{D}_0. \quad (70)$$

Using the inherited pointwise bound $|b_{i,u}| \leq X^{o(1)}$ and $\#\mathcal{A}_s \ll X^{o(1)}Q$, one also has

$$\mathcal{D}_p \ll X^{o(1)}Q \left(1 + \frac{X}{p}\right) \ll X^{o(1)}\frac{QX}{p} \quad (p > y) \quad (71)$$

whenever p occurs in a terminal target.

Proof. The high-prime atom sets are disjoint by [theorem 3.3](#), proving (70). For fixed p , the number of source-row/orbit pairs is bounded by (60); multiplying by the $X^{o(1)}Q$ opposite output rows and the pointwise coefficient bound proves (71). An occurring prime has $p \ll X$, so $1 + X/p \ll X/p$. $\square$

3.2 Balanced and very-high cofactors are complementary

The letter r in (61) is the target cofactor. At the endpoint

$$L_0 = \frac{Q}{D_0} = X^{99979/210000+o(1)}. \quad (72)$$

If $p > \sqrt{N_+}$ and a balanced cofactor descent additionally asks $r \gg_h L_0$, then

$$p = \frac{n_{\rho,j}}{r} \ll \frac{X}{L_0} = X^{110021/210000+o(1)}. \quad (73)$$

This interval is nonempty because $110021/210000 > 1/2$, but it lies far below the very-high cutoff (64). Conversely, for $p > y$,

$$r \leq \frac{N_+}{y} = X^{133/400-\eta+o(1)} = o(L_0). \quad (74)$$

Thus the balanced condition $p > \sqrt{N_+}$, $r \gg_h L_0$ and the very-high condition $p > y$ cannot hold simultaneously for large X . They are complementary coefficient sectors, not two hypotheses that may be combined on one atom.

4 Exact cofactor descent on a global prime band

We begin with a prime band defined from the whole equality packet, not from one selected row. This distinction is necessary because the sign attached to a prime is global: the same prime cannot be frozen in one occurrence and left random in another. In this section and in the determinant-ladder discussion we denote a band prime by q ; the generic incidence and Gram sections denote the same prime role by p .

Let $\mathfrak{A}_X$ be the set of nonzero equality atoms. An atom has a coordinate i , whose data include an orbit time $j = j(i)$, and a source row $\alpha = (\ell_\alpha, d_\alpha)$. Put

$$n_{\alpha,j} = \ell_\alpha d_\alpha j + h, \quad u_{\alpha,j} = d_\alpha n_{\alpha,j}. \quad (75)$$

On the nonzero support, d_α and $n_{\alpha,j}$ are squarefree and

$$(d_\alpha, n_{\alpha,j}) = (d_\alpha, h) = 1. \quad (76)$$

Use the global target endpoints $N_\pm$ from the actual incidence interface, chosen as the extrema of the nonzero support, and define

$$\ell_+ := \max_{\alpha} \ell_\alpha. \quad (77)$$

Empty packets are harmless, so we suppose $\mathfrak{A}_X \neq \emptyset$. The fixed dyadic supports give

$$N_- = X^{1+o(1)}, \quad N_+ = X^{1+o(1)}, \quad \ell_+ = X^{\lambda_L+o(1)}, \quad \lambda_L = \frac{99979}{210000}. \quad (78)$$

Definition 4.1 (Global cofactor band). Let $R = R_X$ satisfy

$$1 \leq R < \frac{N_-}{\sqrt{N_+}}, \quad (79)$$

and set

$$\mathcal{P}_R := \left\{ q \in \mathbb{P} : \sqrt{N_+} < q \leq \frac{N_-}{R} \right\}. \quad (80)$$

Every sign ε_q , $q \in \mathcal{P}_R$, will be set to the physical value -1 in every occurrence of q .

If $R = X^{\theta+o(1)}$, $0 \leq \theta < 1/2$, the band is asymptotically nonempty and has endpoint exponents $1/2$ and $1 - \theta$. The clean balanced choice $R = \ell_+$ will be denoted by

$$\mathcal{P}_{\text{bal}} := \mathcal{P}_{\ell_+}. \quad (81)$$

For this choice the endpoint exponents are

$$\frac{1}{2} = \frac{105000}{210000}, \quad 1 - \lambda_L = \frac{110021}{210000}; \quad (82)$$

their gap is

$$\frac{5021}{210000} > 0. \quad (83)$$

Constants in the dyadic endpoints cannot close this fixed power gap.

Lemma 4.2 (Unique large prime and controlled cofactor). *Suppose that $q \in \mathcal{P}_R$ divides a nonzero target $n_{\alpha,j}$, and write*

$$n_{\alpha,j} = qr. \quad (84)$$

Then q is the only prime from $\mathcal{P}_R$ dividing that target, and

$$R \leq r < \sqrt{N_+}. \quad (85)$$

Moreover

$$s := d_\alpha r \quad (86)$$

is squarefree, $(q, s) = 1$, and

$$u_{\alpha,j} = qs. \quad (87)$$

Proof. Two primes larger than $\sqrt{N_+}$ have product larger than every target in the packet, which proves uniqueness. The definition of the upper endpoint of $\mathcal{P}_R$ gives

$$r = \frac{n_{\alpha,j}}{q} \geq \frac{N_-}{q} \geq R,$$

whereas $q > \sqrt{N_+}$ and $n_{\alpha,j} \leq N_+$ give the strict upper bound in (85). Squarefreeness and (76) show that q, r, d_α are pairwise coprime and squarefree. The last two assertions follow. $\square$

Since $d_+ = X^{10049/52500+o(1)}$ whereas $\sqrt{N_+} = X^{1/2+o(1)}$, one has

$$d_+ < \sqrt{N_+} \quad (88)$$

for all sufficiently large X . Hence no prime in $\mathcal{P}_R$ divides a source factor d_α ; band incidence in the fused label is exactly the target incidence in $n_{\alpha,j}$.

For the chosen R , define each nonzero atom's descended label and deterministic descent sign by

$$(w_{\alpha,j}, \eta_{\alpha,j}) := \begin{cases} (u_{\alpha,j}/q, -1), & q \in \mathcal{P}_R, q \mid n_{\alpha,j}, \\ (u_{\alpha,j}, 1), & \text{if no prime of } \mathcal{P}_R \text{ divides } n_{\alpha,j}. \end{cases} \quad (89)$$

The first case is unambiguous by lemma 4.2, and there $w_{\alpha,j} = s = d_\alpha r$.

Proposition 4.3 (Exact global-band character descent). *Let ε denote the signs outside $\mathcal{P}_R$. For every nonzero equality atom,*

$$\boxed{\chi_{u_{\alpha,j}}(-\mathbf{1}_{\mathcal{P}_R}, \varepsilon) = \eta_{\alpha,j} \chi_{w_{\alpha,j}}(\varepsilon).} \quad (90)$$

At the all-minus point of the remaining signs,

$$\eta_{\alpha,j} \mu(w_{\alpha,j}) = \mu(u_{\alpha,j}). \quad (91)$$

Thus the descended packet has exactly the original physical Möbius corner.

Proof. Untouched atoms give both identities trivially. On a touched atom, $u = qs$, so

$$\chi_u(-\mathbf{1}_{\mathcal{P}_R}, \varepsilon) = -\chi_s(\varepsilon).$$

This is (90). Since q and s are coprime and squarefree,

$$-\mu(s) = \mu(q)\mu(s) = \mu(qs) = \mu(u),$$

which proves (91). $\square$

The use of the full-support quantities N_- and R in (80) is essential. If a prime were selected because one of its occurrences happened to have $r \geq L_0$, another occurrence of the same global prime could have a smaller cofactor. The uniform inequality $r \geq R$ for every occurrence is what makes the compression in the next section valid for the whole packet. The clean choice $R = \ell_+$ is the first scale at which the fiber cost below is subpolynomial without additional cancellation.

5 Cofactor-band compression and the balanced clean range

We now prove a quantitative fiber theorem for the full family $\mathcal{P}_R$. The balanced choice $R = \ell_+$ is its subpolynomial clean case.

Let

$$d_- := \min_{\alpha} d_{\alpha}, \quad d_+ := \max_{\alpha} d_{\alpha}. \quad (92)$$

The endpoint scales imply, for all sufficiently large X ,

$$d_+ \sqrt{N_+} < d_- N_-. \quad (93)$$

Theorem 5.1 (Cofactor-band fiber compression). *Fix one output coordinate i , hence one orbit time $j = j(i)$, and one squarefree descended label w . Then*

$$\#\{\alpha : (i, \alpha) \in \mathfrak{A}_X, w_{\alpha,j} = w\} \leq C_h \tau(w) \left(1 + \frac{L_0}{R}\right). \quad (94)$$

Consequently, if $R = X^{\theta+o(1)}$,

$$\begin{aligned} \rho_R &:= \max_{i,w} \#\{\alpha : w_{\alpha,j(i)} = w\} \\ &\leq \left(1 + \frac{L_0}{R}\right) \exp\left(O\left(\frac{\log X}{\log \log X}\right)\right) = X^{(\lambda_L - \theta)_+ + o(1)}. \end{aligned} \quad (95)$$

In particular $R = \ell_+$ gives $\rho_R = X^{o(1)}$.

Proof. Two untouched rows with the same descended label have the same original fused label. Equality-alias injectivity then makes the rows equal. A touched label is at most $d_+\sqrt{N_+}$, whereas an untouched label is at least d_-N_- . Thus (93) rules out a mixed collision.

It remains to count touched rows with a common label $w = s$. Such a row has

$$d \mid s, \quad r = s/d, \quad rq - dj\ell = h, \quad (96)$$

where $q \in \mathcal{P}_R$. Fix $d \mid s$, hence $r = s/d$, and put $g = (r, dj)$. There is no solution unless $g \mid h$. Given one integral solution (q_0, ℓ_0) , all integral solutions are

$$q = q_0 + \frac{dj}{g}t, \quad \ell = \ell_0 + \frac{r}{g}t, \quad t \in \mathbb{Z}. \quad (97)$$

The source-prime interval has length $O(L_0)$. Because $g \mid h$ and $r \geq R$, it contains at most

$$O\left(1 + \frac{gL_0}{r}\right) = O_h\left(1 + \frac{L_0}{R}\right) \quad (98)$$

parameters t . Primality, support, and mask conditions only delete solutions. Summing over $d \mid s$ proves (94). Since $s \leq d_+\sqrt{N_+} = X^{O(1)}$, the uniform divisor bound and $L_0 = X^{\lambda_L + o(1)}$ prove (95). $\square$

Write the equality coordinate before descent as

$$S_i(\varepsilon) = \sum_{\alpha: (i, \alpha) \in \mathfrak{A}_X} b_{i, \alpha} \chi_{u_{\alpha, j(i)}}(\varepsilon), \quad Z_0(\varepsilon) = \sum_i |S_i(\varepsilon)|^2, \quad (99)$$

and put

$$\mathcal{D}_0 = \sum_i \sum_{\alpha} |b_{i, \alpha}|^2. \quad (100)$$

After freezing $\mathcal{P}_R$, group equal descended labels:

$$\tilde{b}_{i, w} := \sum_{\alpha: w_{\alpha, j(i)} = w} \eta_{\alpha, j(i)} b_{i, \alpha}, \quad \tilde{S}_i(\varepsilon) := \sum_w \tilde{b}_{i, w} \chi_w(\varepsilon). \quad (101)$$

Corollary 5.2 (Exact grouped packet and diagonal cost). *One has*

$$Z_0(-\mathbf{1}_{\mathcal{P}_R}, \varepsilon) = \sum_i |\tilde{S}_i(\varepsilon)|^2. \quad (102)$$

The grouped labels are distinct in each coordinate, and

$$\tilde{\mathcal{D}} := \sum_i \sum_w |\tilde{b}_{i, w}|^2 \leq \rho_R \mathcal{D}_0 \leq X^{(\lambda_L - \theta)_+ + o(1)} \mathcal{D}_0, \quad (103)$$

$$\mathbb{E}_{\varepsilon_p: p \notin \mathcal{P}_R} Z_0(-\mathbf{1}_{\mathcal{P}_R}, \varepsilon) = \tilde{\mathcal{D}}. \quad (104)$$

Proof. The pointwise identity is proposition 4.3 followed by grouping. On each fiber, Cauchy–Schwarz gives

$$|\tilde{b}_{i, w}|^2 \leq \rho_R \sum_{\alpha: w_{\alpha, j(i)} = w} |b_{i, \alpha}|^2.$$

Sum this inequality and use theorem 5.1. Orthogonality of the now distinct Walsh labels proves the expectation. $\square$

At $K_B = X^{1/400 + o(1)}$, Markov's inequality gives

$$\mathbb{P}_{\varepsilon_p: p \notin \mathcal{P}_R} (Z_0(-\mathbf{1}_{\mathcal{P}_R}, \varepsilon) > K_B \mathcal{D}_0) \leq X^{(\lambda_L - \theta)_+ - 1/400 + o(1)}. \quad (105)$$

For the clean choice $R = \ell_+$, a density-one set of the remaining prime environments is therefore good after the entire polynomial-width band $\mathcal{P}_{\text{bal}}$ has been set to its physical value. More generally, (105) is nontrivial whenever $(\lambda_L - \theta)_+ < 1/400$. It does not set the remaining primes to their physical values.

5.1 Residual label exponents passed to robust freezing

The descended labels of touched atoms satisfy

$$w = s = dr \leq d_+ \sqrt{N_+} = X^{\lambda_1 + o(1)}, \quad \lambda_1 = \frac{10049}{52500} + \frac{1}{2} = \frac{36299}{52500}. \quad (106)$$

For

$$z_1 = (\log X)^{1003/1000}, \quad (107)$$

the Rankin projection count for touched labels alone is

$$\nu_{\text{touch}, z_1} \leq X^{\lambda_1(1-1000/1003)+o(1)} = X^{36299/17552500+o(1)}. \quad (108)$$

The exact endpoint margin is

$$\frac{1}{400} - \frac{36299}{17552500} = \frac{30329}{70210000} > 0. \quad (109)$$

We record only the budget interface here. If $R = X^{\theta+o(1)}$, the full transformed packet can fit a Rankin cutoff $z = (\log X)^A$ only when

$$(\lambda_L - \theta)_+ + \frac{62549}{52500} \left(1 - \frac{1}{A}\right) < \frac{1}{400}. \quad (110)$$

For the touched residual sector, the corresponding sufficient budget is

$$(\lambda_L - \theta)_+ + \frac{36299}{52500} \left(1 - \frac{1}{A}\right) < \frac{1}{400}. \quad (111)$$

At the clean choice $\theta = \lambda_L$, $A = 1003/1000$ satisfies the touched condition by (109). It does not satisfy the full condition: untouched labels still have size up to $X^{62549/52500+o(1)}$, whose $A = 1003/1000$ projection exponent is $62549/17552500 > 1/400$. Thus $1003/1000$ belongs only to the touched residual sector. The full probability statement is deferred to the residual-freezing section.

6 Robust freezing after cofactor descent

We now combine the exact band descent with the face argument of TPC-44. The proof is included because the diagonal loss ρ_R creates a useful quantitative tradeoff between the width of the physical band and the logarithmic low-prime cutoff. Write the inherited endpoint budget as

$$K_B = X^{1/400+\kappa_X}, \quad \kappa_X = o(1). \quad (112)$$

Let $\tilde{S}_i = \sum_w \tilde{b}_{i,w} \chi_w$ be the grouped packet in (101); none of its labels is divisible by a prime in $\mathcal{P}_R$. Fix $z < \sqrt{N_+}$, so the low cutoff and the band are disjoint. Split every descended label uniquely as

$$w = ab, \quad P^+(a) \leq z < P^-(b), \quad (113)$$

and let $\tilde{\mathcal{A}}_z$ be the set of low parts a that occur. Put

$$\tilde{\nu}_z := |\tilde{\mathcal{A}}_z|. \quad (114)$$

For a remaining high-prime environment $\xi = (\varepsilon_p)_{p>z, p \notin \mathcal{P}_R}$, define

$$H_{i,a}(\xi) := \sum_{w \leq z=a} \tilde{b}_{i,w} \chi_{w>z}(\xi), \quad W_z(\xi) := \sum_i \sum_{a \in \tilde{\mathcal{A}}_z} |H_{i,a}(\xi)|^2. \quad (115)$$

Lemma 6.1 (Robust descended face). *For every $L \geq 1$,*

$$\mathbb{P}_\xi \left(\sup_{\sigma_{p \leq z}} Z_0(-\mathbf{1}_{\mathcal{P}_R}, \sigma, \xi) > L \tilde{\nu}_z \tilde{\mathcal{D}} \right) \leq L^{-1}. \quad (116)$$

Proof. Distinct descended labels with the same low part have distinct high parts. Walsh orthogonality therefore gives

$$\mathbb{E}_\xi W_z(\xi) = \tilde{\mathcal{D}}. \quad (117)$$

For every assignment σ below z ,

$$\tilde{S}_i(\sigma, \xi) = \sum_{a \in \tilde{\mathcal{A}}_z} \chi_a(\sigma) H_{i,a}(\xi). \quad (118)$$

Cauchy–Schwarz and then summation in i give

$$\sup_{\sigma} Z_0(-\mathbf{1}_{\mathcal{P}_R}, \sigma, \xi) \leq \tilde{\nu}_z W_z(\xi). \quad (119)$$

Markov’s inequality and (117) prove the claim. $\square$

Every untouched descended label is an original label of size at most

$$X^{\lambda_0 + o(1)}, \quad \lambda_0 = 1 + \frac{10049}{52500} = \frac{62549}{52500}, \quad (120)$$

while touched labels are shorter by (106). The standard Rankin argument for squarefree z -smooth integers gives, when $z = (\log X)^A$ and $A > 1$,

$$\tilde{\nu}_z \leq X^{\lambda_0(1-1/A) + o(1)}. \quad (121)$$

For completeness, choose $s = 1 - 1/A + (\log \log X)^{-1/2}$ and bound the number of possible low parts by

$$X^{(\lambda_0 + o(1))s} \prod_{p \leq z} (1 + p^{-s}). \quad (122)$$

The logarithm of the product is $O_A(z^{1-s}) = o(\log X)$, which proves (121).

Theorem 6.2 (Band/cutoff tradeoff). *Suppose*

$$R = X^{\theta + o(1)}, \quad 0 \leq \theta < \frac{1}{2}, \quad z = (\log X)^A, \quad A > 1, \quad (123)$$

and

$$\boxed{(\lambda_L - \theta)_+ + \lambda_0 \left(1 - \frac{1}{A}\right) < \frac{1}{400}.} \quad (124)$$

Let

$$L_X := \exp \left(\frac{\log X}{\sqrt{\log \log X}} \right). \quad (125)$$

Then, for all sufficiently large X ,

$$\mathbb{P}_\xi \left(\sup_{\sigma_{p \leq z}} Z_0(-\mathbf{1}_{\mathcal{P}_R}, \sigma, \xi) > K_B \mathcal{D}_0 \right) \leq L_X^{-1} = o(1). \quad (126)$$

Thus a density-one set of the remaining environments tolerates, at once, the physical assignment on the whole band $\mathcal{P}_R$ and every assignment below z .

Proof. By [corollary 5.2](#),

$$\tilde{\mathcal{D}} \leq X^{(\lambda_L - \theta)_+ + o(1)} \mathcal{D}_0. \quad (127)$$

Combine this with (121). Choose $\delta > 0$ so that the left side of (124) is at most $1/400 - 2\delta$. The strict gap absorbs both the $X^{o(1)}$ factors and $L_X = X^{o(1)}$, giving

$$L_X \tilde{\nu}_z \tilde{\mathcal{D}} \leq X^{1/400 - \delta} \mathcal{D}_0 \leq K_B \mathcal{D}_0 \quad (128)$$

for sufficiently large X , with the inherited harmless subpower convention in K_B . Apply [lemma 6.1](#) with $L = L_X$. $\square$

Corollary 6.3 (Clean balanced freezing with the legacy cutoff). *Take $R = \ell_+$, $A = 501/500$, and $z = (\log X)^{501/500}$. Then*

$$\lambda_0 \left(1 - \frac{500}{501}\right) = \frac{62549}{26302500}, \quad \frac{1}{400} - \frac{62549}{26302500} = \frac{12829}{105210000} > 0. \quad (129)$$

Consequently (126) holds while the entire band

$$\sqrt{N_+} < p \leq N_-/\ell_+ \quad (130)$$

is fixed to the physical sign -1 .

The inequality is open, so a small part of the endpoint budget can be spent to move beyond the clean cofactor scale while keeping the same low cutoff.

Corollary 6.4 (A budgeted extension beyond balance). *Let*

$$R = \ell_+ X^{-1/10000}, \quad A = \frac{501}{500}. \quad (131)$$

Then the total power cost and its remaining margin are

$$\frac{1}{10000} + \frac{62549}{26302500} = \frac{260717}{105210000}, \quad (132)$$

$$\frac{1}{400} - \frac{260717}{105210000} = \frac{577}{26302500} > 0. \quad (133)$$

Hence the old low-prime cutoff remains valid and the physically frozen band extends to exponent

$$1 - \lambda_L + \frac{1}{10000} = \frac{55021}{105000} = 0.5240095238 \dots \quad (134)$$

One may trade more of the low cutoff for a wider band. For example,

$$R = \ell_+ X^{-1/1000}, \quad A = \frac{1001}{1000} \quad (135)$$

has exact margin

$$\frac{1}{400} - \frac{1}{1000} - \frac{62549}{52552500} = \frac{65119}{210210000} > 0, \quad (136)$$

and reaches band exponent $110231/210000 = 0.5249095238 \dots$. These points illustrate one continuous frontier; neither is asserted to be optimal for the actual coefficients.

6.1 The shortened touched component

Let Z_{touch} be the equality energy formed only from atoms touched by $\mathcal{P}_{\text{bal}}$, after the band signs have been set to -1 and equal residual labels have been grouped. This is a standalone nonnegative Walsh packet. Its labels have exponent $\lambda_1 = 36299/52500$ and its grouped diagonal is at most $X^{o(1)}\mathcal{D}_{\text{touch}}$, where $\mathcal{D}_{\text{touch}}$ is its original atomic diagonal.

Corollary 6.5 (Improved cutoff for the touched packet). *For*

$$z_1 = (\log X)^{1003/1000}, \quad A_1^* := \left(1 - \frac{1}{400\lambda_1}\right)^{-1} = \frac{145196}{144671}, \quad (137)$$

one has $1 < 1003/1000 < A_1^*$ and

$$\frac{1}{400} - \lambda_1 \left(1 - \frac{1000}{1003}\right) = \frac{30329}{70210000} > 0. \quad (138)$$

Consequently a density-one set of the residual high-prime environments controls every assignment below z_1 for Z_{touch} within the $K_B\mathcal{D}_{\text{touch}}$ scale.

This last corollary is intentionally componentwise. The untouched column still has labels of exponent λ_0 , and the norm of the sum of touched and untouched columns contains cross terms. Therefore $(\log X)^{1003/1000}$ is not a full-packet cutoff.

Finally, the failure probability L_X^{-1} is summable on dyadic scales. Prescribe one inherited packet and one admissible pair (R_X, z_X) at each $X = 2^k$, and couple the remaining signs by a single global independent prime-sign field. The first Borel–Cantelli lemma makes the corresponding tradeoff assertion hold at all sufficiently large dyadic scales almost surely. The same is true for any fixed finite family of such prescribed choices after a union bound. No independence between scales is required; no uniformity over all packet or parameter sequences is claimed.

7 Two exact very-high-prime Gram forms

There are two natural operations on the signs below the cutoff y : one may set them to their physical value -1 , or one may average them. The first operation retains every cross-label inner product and gives an affine polynomial in the high signs. The second keeps only equal residual-label fibers and gives a pure quadratic Gram synthesis. These are different identities in different Hilbert spaces.

Let $\mathcal{P}_y$ be the finite set of primes $p > y$ occurring as the high part of an actual equality label. Use the factorization

$$u = sq, \quad P^+(s) \leq y, \quad q = 1 \text{ or } q \in \mathcal{P}_y \quad (139)$$

from [theorem 3.3](#).

7.1 Physical low signs: an affine output-space form

Define vectors in the original output space $\mathcal{H}_{\text{out}}$ by

$$(v_0)_i := \sum_{\substack{u \in \mathcal{U}_i \\ u > y=1}} \mu(u)b_{i,u}, \quad (140)$$

$$(v_p)_i := \sum_{\substack{u \in \mathcal{U}_i \\ u > y=p}} \mu(u/p)b_{i,u}, \quad p \in \mathcal{P}_y. \quad (141)$$

By the singleton assertion in [theorem 3.3](#), the sum in (141) has at most one term for each fixed i .

Theorem 7.1 (Physical-low affine normal form). *After setting every sign at $q \leq y$ to -1 , the complete equality column is*

$$\boxed{S(-\mathbf{1}_{\leq y}, \varepsilon_{> y}) = v_0 + \sum_{p \in \mathcal{P}_y} \varepsilon_p v_p} \quad \text{in } \mathcal{H}_{\text{out}}. \quad (142)$$

Consequently,

$$\boxed{Z_0(-\mathbf{1}_{\leq y}, \varepsilon_{> y}) = \left\| v_0 + \sum_{p \in \mathcal{P}_y} \varepsilon_p v_p \right\|^2}. \quad (143)$$

Its high-prime Walsh spectrum has degree at most two. Explicitly, for the centered polynomial $F = Z_0 - \mathcal{D}_0$, the coefficients after the low restriction are

$$c_\emptyset = \|v_0\|^2 + \sum_p \|v_p\|^2 - \mathcal{D}_0, \quad (144)$$

$$c_p = 2 \operatorname{Re} \langle v_0, v_p \rangle, \quad (145)$$

$$c_{pq} = 2 \operatorname{Re} \langle v_p, v_q \rangle \quad (p < q), \quad (146)$$

and every coefficient of degree at least three is zero. At the physical high corner,

$$\boxed{Z_0(-\mathbf{1}) = \left\| v_0 - \sum_{p \in \mathcal{P}_y} v_p \right\|^2}. \quad (147)$$

Proof. If $u_{> y} = 1$, its character at the restricted low face is $\mu(u)$. If $u = sp$, the same character is $\mu(s)\varepsilon_p$. Grouping the coordinate formulas equations (34) and (35) gives (142); taking the output norm gives (143). Expansion of that norm proves equations (144) to (146). Finally put every remaining sign equal to -1 . $\square$

The formula is literal in the row variables. If the unique left source row at (j, p) is $\alpha_p(j)$ and

$$n_{\alpha_p(j), j} = pr_p(j), \quad (148)$$

then

$$(v_p)_{L, \gamma, j} = \mu(d_{\alpha_p(j)} r_p(j)) B_{\alpha_p(j); \gamma, j}^L, \quad (149)$$

with value zero when no such row exists. Similarly, if the unique right summation row is $\gamma_p(j)$, then

$$(v_p)_{R, \alpha, j} = \mu(d_{\gamma_p(j)} r_p(j)) B_{\gamma_p(j); \alpha, j}^R. \quad (150)$$

Thus the residual physical sign is precisely $\mu(dr)$. The high prime itself supplies the final minus sign in (147).

For example, the left part of the quadratic coefficient is

$$\begin{aligned} c_{pq}^L &= 2 \operatorname{Re} \sum_{\gamma, j} \mu(d_{\alpha_p(j)} r_p(j)) \mu(d_{\alpha_q(j)} r_q(j)) \\ &\quad \times B_{\alpha_p(j); \gamma, j}^L \overline{B_{\alpha_q(j); \gamma, j}^L}, \end{aligned} \quad (151)$$

where a summand is present only when

$$pr_p(j) - m_{\alpha_p(j)} j = h, \quad qr_q(j) - m_{\alpha_q(j)} j = h. \quad (152)$$

In particular,

$$pr_p(j) - qr_q(j) = (m_{\alpha_p(j)} - m_{\alpha_q(j)}) j. \quad (153)$$

The right contribution is obtained from (150). This is an actual affine determinant sum with the source weights and joint masks retained.

The degree-two reduction also simplifies the TPC-44 coefficient correlation. Process the primes in $\mathcal{P}_y$ increasingly, after all low signs have already been restricted. Just before processing p , put

$$w_p := v_0 - \sum_{\substack{q \in \mathcal{P}_y \\ q < p}} v_q. \quad (154)$$

The surviving coefficients are

$$c_{\emptyset}^{(p)} = \|w_p\|^2 + \sum_{q \geq p} \|v_q\|^2 - \mathcal{D}_0, \quad (155)$$

$$c_q^{(p)} = 2 \operatorname{Re} \langle w_p, v_q \rangle \quad (q \geq p), \quad (156)$$

$$c_{qr}^{(p)} = 2 \operatorname{Re} \langle v_q, v_r \rangle \quad (p \leq q < r). \quad (157)$$

Let $C_{p^-}(m)$ denote the Walsh coefficient of χ_m in this current centered polynomial after all lower signs and all $q \in \mathcal{P}_y$ with $q < p$ have been fixed to -1 ; the index $m = 1$ denotes the constant coefficient $C_{p^-}(1) = c_{\emptyset}^{(p)}$. Hence the general sum over residual kernels is exactly

$$\boxed{\sum_m C_{p^-}(m) \overline{C_{p^-}(pm)} = c_{\emptyset}^{(p)} c_p^{(p)} + \sum_{q > p} c_q^{(p)} c_{pq}.} \quad (158)$$

Indeed, a residual m with at least two prime factors would make pm have degree at least three, whose coefficient vanishes.

7.2 Averaged low signs: the residual-label Gram form

Now use a different Hilbert space,

$$\mathcal{H}_{\text{res}} := \ell^2 \{ (i, s) : i \in \mathcal{I}_L \sqcup \mathcal{I}_R, \ s \text{ squarefree}, \ P^+(s) \leq y \}. \quad (159)$$

For $p \in \mathcal{P}_y$, define

$$(x_p)_{i,s} := \begin{cases} b_{i,sp}, & sp \in \mathcal{U}_i, \\ 0, & sp \notin \mathcal{U}_i. \end{cases} \quad (160)$$

Let

$$\mathcal{D}_H := \sum_{p \in \mathcal{P}_y} \sum_i \sum_{\substack{u \in \mathcal{U}_i \\ u > y = p}} |b_{i,u}|^2. \quad (161)$$

Theorem 7.2 (Low-averaged pure Gram identity). *Assume both cutoff conditions (62) and (63). Then*

$$\boxed{\mathbb{E}_{\varepsilon_q: q \leq y} Z_0(\varepsilon_{\leq y}, -\mathbf{1}_{>y}) = \mathcal{D}_0 - \mathcal{D}_H + \left\| \sum_{p \in \mathcal{P}_y} x_p \right\|_{\mathcal{H}_{\text{res}}}^2.} \quad (162)$$

Moreover,

$$\sum_p \|x_p\|^2 = \mathcal{D}_H, \quad \|x_p\|^2 = \|v_p\|^2, \quad (163)$$

and therefore

$$\mathbb{E}_{\leq y} Z_0(\varepsilon_{\leq y}, -\mathbf{1}_{>y}) = \mathcal{D}_0 + 2 \operatorname{Re} \sum_{p < q} \langle x_p, x_q \rangle. \quad (164)$$

There is no linear term in (164).

Proof. Fix an output coordinate i . A label with no prime above y is its own low Walsh label u , and distinct such full labels are orthogonal under the low-sign average. A label sp with $p > y$ has low Walsh label s , while its fixed high character contributes the sign -1 .

No no-high full label can have the same low label as a high-prime residual. Indeed, by (55), a no-high label has size at least D_-N_- , whereas

$$s = \frac{d_\rho n_{\rho,j}}{p} \leq \frac{D_+N_+}{y} < D_-N_- \quad (165)$$

for a high-prime label. Thus low Walsh orthogonality leaves the atomic energy $\mathcal{D}_0 - \mathcal{D}_H$ from the no-high labels and, for every low label s , the coefficient

$$- \sum_{p \in \mathcal{P}_y} b_{i,sp} \quad (166)$$

from the high-prime labels. Squaring (166) and summing (i, s) gives (162). Unique high-prime factorization partitions the high atoms, proving the first identity in (163). Fixed- (j, p) singleton incidence shows that (141) contains at most one term per output coordinate; multiplication by $\mu(s)$ preserves its modulus, proving the second. Expanding the last norm now gives (164). $\square$

The Möbius factors suppressed in the stripped definition of x_p may be restored without changing the synthesis norm. Indeed, the diagonal map

$$(Uy)_{i,s} := \mu(s)y_{i,s} \quad (167)$$

is unitary on $\mathcal{H}_{\text{res}}$, and on every occurring factorization $s = dr$ one has $\mu(s) = \mu(d)\mu(r)$. Consequently

$$\left\| \sum_p x_p \right\| = \left\| \sum_p Ux_p \right\|. \quad (168)$$

This is the precise bridge from the pure Gram form to the source- and cofactor-Möbius coefficient slots discussed below; it does not impose independence or cancellation on either factor.

The matching condition in this Gram space is completely explicit:

$$\langle x_p, x_q \rangle = \sum_i \sum_{\substack{s \\ sp, sq \in \mathcal{U}_i}} b_{i,sp} \overline{b_{i,sq}}. \quad (169)$$

For a left coordinate, a term in (169) consists of rows α, β and cofactors r_α, r_β satisfying

$$pr_\alpha = \ell_\alpha d_\alpha j + h, \quad qr_\beta = \ell_\beta d_\beta j + h, \quad (170)$$

$$d_\alpha r_\alpha = d_\beta r_\beta = s, \quad (171)$$

and its weight is

$$B_{\alpha;\gamma,j}^L \overline{B_{\beta;\gamma,j}^L}. \quad (172)$$

The right formula exchanges the two row systems exactly as in (44).

Because every active d and r in (171) is squarefree and coprime, one has

$$d_\alpha \mid s, \quad r_\alpha = s/d_\alpha, \quad \ell_\alpha = \frac{ps - hd_\alpha}{d_\alpha^2 j}, \quad (173)$$

and the same formula with (q, d_β) . Thus a fixed (i, s, p) -fiber has at most $\tau(s)$ candidate source divisors; integrality, primality, dyadic support, and the actual mask only delete candidates. This is the exact matching mechanism behind the residual Gram space.

Remark 7.3 (The two Gram matrices must not be identified). The physical-low coefficient $\langle v_p, v_q \rangle_{\mathcal{H}_{\text{out}}}$ sums every pair of p - and q -rows sharing an output coordinate and weights it by $\mu(s_p)\mu(s_q)$. The low-averaged coefficient $\langle x_p, x_q \rangle_{\mathcal{H}_{\text{res}}}$ retains only the exact matches $s_p = s_q$; on such a match the two low Möbius signs multiply to one. Although $\|v_p\| = \|x_p\|$, their cross inner products are different. The affine form (143) and the pure Gram form (162) therefore serve different descent routes.

The remaining estimate in either route is coefficient-specific. In the physical-low form it is an angle bound for the actual vectors v_p , hence for the signed determinant sums (151). In the low-averaged form it is a Bessel bound for the matching synthesis

$$\left\| \sum_{p \in \mathcal{P}_y} x_p \right\|^2 \quad (174)$$

on the ladders (170)–(171). Singleton prime incidence and the diagonal identity (163) do not by themselves control either coherent sum.

8 The unbalanced fixed-determinant ladders

The upper endpoint N_-/ℓ_+ in (81) is the natural boundary of the fiber argument. We now retain only the uniqueness condition $q > \sqrt{N_+}$ and allow the cofactor $r = n/q$ to be smaller than the source-prime scale.

Fix an orbit time j , a residual label s , and a divisor $d \mid s$. Put $r = s/d$. Every row that deletes to s solves

$$rq - dj\ell = h. \quad (175)$$

Let $g = (r, dj)$. If $g \nmid h$, there are no such rows. Otherwise, for any one integral solution (q_0, ℓ_0) , the complete ladder is

$$\boxed{q = q_0 + \frac{dj}{g}t, \quad \ell = \ell_0 + \frac{r}{g}t, \quad t \in \mathbb{Z}.} \quad (176)$$

The two affine forms have fixed nonzero cross determinant

$$\frac{dj}{g}\ell_0 - \frac{r}{g}q_0 = -\frac{h}{g}. \quad (177)$$

In the physical packet both q and ℓ must be prime, in their respective support windows. This primality condition deletes points from (176); it does not shorten the ambient ladder available to a coefficient-blind argument.

Proposition 8.1 (Dyadic residual-fiber capacity). *Let $\mathcal{R} \subset [R, 2R]$ be a dyadic cofactor block. For fixed (j, s) , the number of rows with $r = s/d \in \mathcal{R}$, $q > \sqrt{N_+}$, and descended label s is at most*

$$C_h \sum_{\substack{d \mid s \\ s/d \in \mathcal{R}}} \left(1 + \frac{L_0 d}{s}\right) \leq C_h \tau(s) \left(1 + \frac{L_0}{R}\right). \quad (178)$$

Proof. For fixed d , the step of the ℓ -form in (176) is r/g . Since $g \mid h$ and the source-prime window has length $O(L_0)$, the number of integral parameters is

$$O\left(1 + \frac{gL_0}{r}\right) = O_h\left(1 + \frac{L_0 d}{s}\right).$$

Sum over the eligible divisors $d \mid s$, and use $r \geq R$. □

At the original equality endpoint,

$$L_0 = X^{99979/210000+o(1)}, \quad J = X^{133/400+o(1)}. \quad (179)$$

At the old threshold $q \asymp Q$, where the cofactor scale is $R \asymp J$, the unsieved ladder capacity is already

$$\frac{L_0}{J} = X^{15077/105000+o(1)}. \quad (180)$$

It exceeds the endpoint allowance $K_B = X^{1/400+o(1)}$ by the fixed power

$$\frac{15077}{105000} - \frac{1}{400} = \frac{29629}{210000} > 0. \quad (181)$$

For $r = O(1)$, the ambient ladder has length $X^{99979/210000+o(1)}$.

The parametrization also explains why a standard upper-bound sieve is not the missing theorem. It may remove logarithmic powers from the number of parameters for which both affine forms are prime, but the power-sized factor L_0/R remains. More importantly, the physical coefficients on the surviving parameters are neither identical to an unweighted prime-pair count nor controlled by their cardinality. The needed input is a weighted Bessel or dispersion estimate for the actual source and mask coefficients along (176).

Equation (175) is a literal fixed-determinant equation, not a heuristic analogy. Nevertheless, a theorem for two arbitrary coefficient sequences and otherwise smooth weights cannot be invoked after silently inserting, in addition, the prime restrictions on q, ℓ , the Möbius cofactor weight, and the moving source/mask coefficient. Those quantifiers form the next arithmetic interface.

9 Sharp operator boundaries and the actual Walsh spectrum

The fiber loss in the preceding sections is an exact operator norm, not an artifact of Cauchy–Schwarz. We first record the finite-dimensional statement.

Let E be a finite atom set, let $\pi : E \rightarrow W$ be a label map, and define

$$(G_\pi x)(w) := \sum_{e:\pi(e)=w} x(e). \quad (182)$$

Proposition 9.1 (Sharp grouping norm). *If*

$$\rho(\pi) := \max_w |\pi^{-1}(w)|, \quad (183)$$

then

$$\|G_\pi\|_{\ell^2(E) \rightarrow \ell^2(W)}^2 = \rho(\pi). \quad (184)$$

Proof. Cauchy–Schwarz on each fiber gives the upper bound. On a fiber of maximal cardinality, take x constant and nonzero and set it to zero off that fiber. The quotient of the squared norms is exactly the fiber cardinality. $\square$

For large-prime cofactor descent, all atoms in one descended fiber have the same physical Möbius phase. Indeed, if $s = dr$, then

$$\mu(d)\mu(qr) = -\mu(dr) = -\mu(s). \quad (185)$$

Thus Möbius does not cancel different points of a fixed-determinant ladder after their common large prime has been removed. Any power improvement over the support exponent in (178), beyond possible logarithmic thinning from primality, must use the actual complex source/mask coefficients or cancellation between different residual labels s ; it cannot arise from independent Möbius signs inside one fiber.

9.1 What is proved directly about $C_X(\kappa)$

Let $C_X(\kappa)$ be the actual Walsh coefficient defined by

$$Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\kappa > 1} C_X(\kappa) \chi_\kappa(\varepsilon). \quad (186)$$

Every equality label contains at most one prime from $\mathcal{P}_{\text{bal}}$. Therefore

$$C_X(\kappa) = 0 \quad \text{if } \kappa \text{ contains at least three primes from } \mathcal{P}_{\text{bal}}. \quad (187)$$

There is also no pure singleton coefficient:

$$C_X(q) = 0 \quad (q \in \mathcal{P}_{\text{bal}}). \quad (188)$$

Indeed, a pair kernel equal to q would require an untouched label $u \geq d_- N_-$ to equal a touched label with q deleted, which is at most $d_+ \sqrt{N_+}$. This contradicts (93).

For distinct $p, q \in \mathcal{P}_{\text{bal}}$, a pure coefficient $C_X(pq)$ can occur only when the two labels delete to the same residual label s . Introduce the Hilbert space with coordinates (i, s) , and let $x_q(i, s)$ be the coefficient of the unique row carrying q and deleting to s , with value zero when that row is absent. Uniqueness follows because two such rows would have the same original label qs in one coordinate, contrary to lemma 2.1. Then

$$C_X(pq) = 2 \operatorname{Re} \langle x_p, x_q \rangle, \quad (189)$$

$$\mathcal{D}_{\text{bal}} := \sum_{q \in \mathcal{P}_{\text{bal}}} \|x_q\|^2 \leq \mathcal{D}_0. \quad (190)$$

The second line is the atomic diagonal of the band-touched atoms: equality-label injectivity makes each fixed (i, s, q) entry a singleton, and the same atom cannot carry two band primes.

Averaging every prime sign outside the band leaves the exact quadratic projection

$$\boxed{\mathbb{E}_{\varepsilon_a: a \notin \mathcal{P}_{\text{bal}}} Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\substack{p < q \\ p, q \in \mathcal{P}_{\text{bal}}}} C_X(pq) \varepsilon_p \varepsilon_q.} \quad (191)$$

At the physical band point this is equivalently

$$\mathbb{E}_{a \notin \mathcal{P}_{\text{bal}}} Z_0(-\mathbf{1}_{\mathcal{P}_{\text{bal}}}, \varepsilon) = \mathcal{D}_0 - \mathcal{D}_{\text{bal}} + \left\| \sum_{q \in \mathcal{P}_{\text{bal}}} x_q \right\|^2. \quad (192)$$

Combining this identity with (104) proves the actual-coefficient estimate

$$\mathcal{D}_0 + \sum_{\substack{p < q \\ p, q \in \mathcal{P}_{\text{bal}}}} C_X(pq) \leq X^{o(1)} \mathcal{D}_0. \quad (193)$$

This controls a genuine polynomial band of the physical rough spectrum; it is not an estimate for every kernel containing a band prime.

9.2 The sharp continuation boundary

If the upper endpoint of $\mathcal{P}_{\text{bal}}$ is crossed, cofactors $r < L_0$ appear and the exact norm in proposition 9.1 may cost L_0/r . If a physical fiber contains H atoms, coefficients of equal phase supported on that fiber give grouped energy H times their atomic energy. This is a sharp statement conditional on the declared support; it neither asserts that the actual masks choose equal phases nor proves that a fiber of polynomial cardinality occurs.

Accordingly, the results above do not show that the residual TPC-44 correlation product is small for all high primes, and they do not show that it is large. They prove a physical freezing theorem on the balanced band and identify the first adjacent range where support and diagonal information cease to suffice. No fixed-shift Chowla estimate, prime-pair asymptotic, parity breach, or twin-prime conclusion follows.

10 The present analytic boundary

The congruence and determinant reformulations above place the remaining TPC term near several established dispersion and Kloosterman–fraction estimates. The resemblance is exact at the level of the equation, but not at the level of the coefficient class or of the available exponent range. We record the distinction explicitly. In this section Q_{mod} denotes a modulus scale; it is not the row-length parameter used elsewhere in the paper.

10.1 Unbalanced-convolution dispersion

Fouvry and Radziwiłł prove distribution estimates for a convolution $\sum_{mn \equiv a \pmod{q}} \alpha_m \beta_n$ averaged over $q \asymp Q_{\text{mod}}$, with divisor-bounded coefficients and a Siegel–Walfisz hypothesis on the short sequence [4, Corollary 1.1]. With $MN \asymp X$, their first range is

$$\exp((\log X)^\epsilon) \leq N \leq Q_{\text{mod}}^{-11/12} X^{17/36-\epsilon}. \quad (194)$$

Their two other stated ranges have, respectively,

$$\begin{aligned} N &\leq X^{7/90-\epsilon}, & Q_{\text{mod}} &\leq X^{53/105-\epsilon}, \\ N &\leq X^{101/630-\epsilon}, & Q_{\text{mod}} &\leq X^{53/105-\epsilon}, \end{aligned} \quad (195)$$

with the further hypotheses specified there. Their prime-modulus variant removes the Siegel–Walfisz input but retains the power range in (194) [4, Theorem 1.2].

The algebraic match with our incidence relation is literal:

$$p \mid (mj + h) \iff mj \equiv -h \pmod{p}.$$

The parameter match fails, however. At the first large-prime block $p \asymp Q_{\text{mod}} = X^{1/2}$, (194) permits only

$$N \leq X^{17/36-11/24-\epsilon} = X^{1/72-\epsilon},$$

whereas the TPC orbit variable has length

$$J = X^{133/400+o(1)}.$$

The exponent deficit is

$$\frac{133}{400} - \frac{1}{72} = \frac{1147}{3600}. \quad (196)$$

The alternatives in (195) are also shorter than J . Moreover, successive physical prime restrictions produce coefficient fields depending on the preceding restrictions, rather than a single fixed product $\alpha_m \beta_j$. Thus the theorem has the right congruence but neither the required length nor the required restriction-stable coefficient interface.

Wright’s partially fixed-modulus Kloosterman estimate improves the first unbalanced range to

$$N \leq Q_{\text{mod}}^{-33/28} X^{17/28-\epsilon} \quad (197)$$

and improves the accompanying modulus cap from $53/105$ to $45/89$ in the other two cases [7, Corollary 2.2]. At $Q_{\text{mod}} = X^{1/2}$, (197) becomes

$$N \leq X^{17/28-33/56-\epsilon} = X^{1/56-\epsilon}.$$

Consequently the gap to the TPC length is still

$$\frac{133}{400} - \frac{1}{56} = \frac{881}{2800}. \quad (198)$$

The advance is genuine, but it does not enter the balanced TPC regime.

10.2 The fixed-determinant coefficient-slot barrier

Bettin and Chandee estimate trilinear forms with Kloosterman fractions for three arbitrary coefficient sequences [1, Theorem 1]. Their fixed-determinant consequence concerns

$$m_1 n_2 - m_2 n_1 = \Delta \tag{199}$$

and places arbitrary arithmetic sequences on n_1, n_2 , while the weights on m_1, m_2 are smooth dyadic functions with controlled derivatives [1, Corollary 1]. Hence this corollary has two, not three, arbitrary arithmetic coefficient slots.

Our cofactor equation

$$pt - mj = h \tag{200}$$

has the same determinant shape. Its actual coefficient system contains the prime restriction in p , the Möbius weight $\mu(t)$, and the source weight $\mu(d_\rho)$ attached to $m_\rho = \ell_\rho d_\rho$, together with the inherited masks in m_ρ . Only the orbit variable j is naturally smooth. Assigning two of p, t, m to the arbitrary slots in (199) therefore leaves the third nonsmooth arithmetic sequence in a slot for which the corollary requires a smooth weight. The three-sequence trilinear theorem does not by itself restore this lost slot in the fixed- h determinant corollary. This is the precise *third-coefficient barrier*; it is not merely a difference of notation.

Remark 10.1 (A diagnostic exponent ledger, not a theorem). It is informative to make a deliberately counterfactual calculation. Suppose that two TPC variables could be replaced by admissible smooth weights and that the remaining arbitrary variables had lengths

$$P = X^{267/400}, \quad T = X^{133/400}, \quad PT = X.$$

Substituting these lengths into the error term of [1, Corollary 1] produces the formal exponent

$$\frac{1}{2} + \frac{7}{20} + \frac{1}{4} \frac{267}{400} = 1 + \frac{27}{1600}.$$

A formal comparison with an $X^{1+1/400+o(1)}$ endpoint budget then leaves $23/1600$ in the exponent. This computation is *not an application* of the Bettin–Chandee corollary: the counterfactual smoothing discards an actual arithmetic weight, and a scalar determinant-count error is not automatically the vector Gram/Bessel energy required here. It is neither a lower bound nor an impossibility theorem. It records only that the most naive two-arbitrary-slot substitution would not close the desired exponent ledger.

10.3 A withdrawn bilinear claim

The first version of the 2026 preprint of Dong, Robles and Zeindler announced an improved bilinear Kloosterman–fraction bound and a saving of $1/12$ over the trivial estimate in the balanced case [3, Theorems 1.4–1.6]. The authors subsequently withdrew the paper. The withdrawal notice states that a missing factor in one equation changes L^5 to L^7 , so the argument no longer yields the claimed improved bound [3]. We therefore do not use the announced $1/12$ saving as a theorem or as an input anywhere in this paper.

There is a second, logically independent boundary. Even if the announced bilinear estimate were available, it concerns a form with two arbitrary sequences and a Kloosterman phase. No fixed- h determinant corollary with the three arithmetic weights in (200) is supplied. Thus that announced estimate would improve a possible analytic subroutine, but would not by itself cross the third-coefficient interface.

10.4 Why the Bourgain–Sarnak–Ziegler criterion is not a black box

The Bourgain–Sarnak–Ziegler criterion gives a powerful route from prime-dilate decorrelation to multiplicative orthogonality. In one form, if $|F| \leq 1$, $|\nu| \leq 1$ is multiplicative, and

$$\left| \sum_{m \leq M} F(p_1 m) \overline{F(p_2 m)} \right| \leq \tau M$$

uniformly for the relevant distinct primes p_1, p_2 , then

$$\left| \sum_{n \leq N} \nu(n) F(n) \right| \leq 2\sqrt{\tau \log(1/\tau)} N$$

for sufficiently large parameters [2, Theorem 2].

This criterion identifies the right qualitative phenomenon, but it is not directly applicable to the TPC coefficient. Here the coefficient is sparse and depends on the scale X ; its normalization is governed by a projective energy rather than by the ambient interval length. More importantly, after a physical restriction at one prime, the coefficient used at the next prime is a new restricted field, so the needed correlations involve restriction-dependent pairs rather than the dilates of one fixed bounded function F . Dividing by an ℓ^∞ norm and applying the criterion would return an ambient-length estimate, not the endpoint Gram-energy bound. A usable BSZ-type reduction would therefore first require a new restriction-stable, energy-normalized prime-dilate theorem; the cited criterion does not supply it.

10.5 Exact frontier statement

The preceding comparisons do not show that (200) is intractable. They show only that the following interface is absent from the cited black boxes:

- (i) the shift h is fixed rather than averaged;
- (ii) the prime, cofactor-Möbius, and source-Möbius/mask weights remain simultaneously present;
- (iii) the estimate is stable under the projective and physical restrictions inherited from the TPC packet; and
- (iv) the output controls a vector-valued Bessel/Gram synthesis norm, not only a scalar congruence discrepancy.

Any future positive input must address these four points explicitly. An estimate averaged over h , or one obtained after replacing a nonsmooth coefficient by a smooth weight, may be valuable in its own right but does not close the fixed physical TPC gate.

11 The next gate

The preceding descent separates three tasks that should not be merged under the phrase “Möbius randomness.” We formulate them at the level of the actual coefficient packet.

11.1 A block synthesis estimate

Let $\mathcal{F}_{i,s}(R)$ denote one residual fiber of atom indices e with cofactor $r \asymp R$, and let $b_{i,e}$ be the inherited complex atom. The support argument gives

$$\sum_{i,s} \left| \sum_{e \in \mathcal{F}_{i,s}(R)} b_{i,e} \right|^2 \ll_h X^{o(1)} \left(1 + \frac{L_0}{R} \right) \sum_{i,e} |b_{i,e}|^2. \quad (201)$$

The squared norm of the exact grouping operator is the actual maximal fiber size. Thus (201) is sharp when its ambient upper bound is replaced by that maximal size; the displayed upper bound need not itself be attained. A power improvement below the actual support frontier must therefore use the source phases, opposite-row masks, or cancellation between different residual labels.

The natural positive target is a coefficient-specific replacement

$$\sum_{i,s} \left| \sum_{e \in \mathcal{F}_{i,s}(R)} b_{i,e} \right|^2 \ll X^{o(1)} \mathcal{B}(R) \sum_{i,e} |b_{i,e}|^2, \quad (202)$$

where $\mathcal{B}(R)$ is strictly smaller than $1 + L_0/R$ on at least one new cofactor range. Even a small power improvement would enlarge the physical band in section 6. Estimate (202) is a target, not an assumption used in any unconditional theorem above.

11.2 The very-large-prime Bessel gate

For the low-averaged rough projection, the exact missing estimate is

$$\left\| \sum_{p > y} x_p \right\|^2 \ll X^{o(1)} K_B \sum_{p > y} \|x_p\|^2. \quad (203)$$

Together with the identity in section 7, this would place that projected sector within the endpoint budget. It would not by itself fix the signs below y . For the complete physical affine face, a sufficient pair of estimates is

$$\|v_0\|^2 \ll X^{o(1)} K_B \mathcal{D}_0, \quad (204)$$

$$\left\| \sum_{p > y} v_p \right\|^2 \ll X^{o(1)} K_B \mathcal{D}_0. \quad (205)$$

Indeed the triangle inequality then bounds $\|v_0 - \sum_p v_p\|^2$. The first line is a base-sector problem and the second is the physical high-prime synthesis problem. Neither follows from the pure rough projection, because lower-sign averaging deletes their cross terms.

11.3 The required determinant input

After dyadic decomposition in the cofactor and source variables, a candidate analytic theorem would have to estimate the fixed equation

$$rp - dj\ell = h \quad (206)$$

with all of the following features retained:

- (i) p and ℓ are prime;

- (ii) the cofactor and source-divisor signs contribute Möbius weights rather than smooth cutoffs;
- (iii) the actual projective mask remains in the coefficient;
- (iv) h is fixed;
- (v) the estimate is vector-valued and stable under the preceding physical prime restrictions;
and
- (vi) the final normalization is the atomic energy, not the ambient number of integer solutions.

The Bettin–Chandee fixed-determinant corollary reviewed here lacks one of these arithmetic coefficient slots; see [section 10](#). Proving a theorem with all six features is the direct route to (202). Proving it after averaging h , smoothing one arithmetic slot, or restricting to a favorable cofactor block would still be a meaningful weaker theorem, but it would have to be labelled accordingly.

11.4 A reversible research program

The exact descent leaves several logically independent doors.

- Door A.** Prove a power saving over (201) on one additional cofactor block and enlarge the unconditional physical band.
- Door B.** Establish a three-arithmetic-weight, fixed- h determinant estimate strong enough for (203).
- Door C.** First prove an averaged-shift or smoothed-slot analogue, thereby identifying which of fixed h , the third coefficient, or the vector-energy normalization is the dominant obstruction.
- Door D.** Construct an actual-mask coefficient packet saturating the support norm. Such a result would be a sharp negative boundary for support-only methods, without asserting that the physical TPC coefficients themselves saturate it.

The current paper completes the geometric reduction needed before any of these analytic attacks. Failure at one door does not invalidate the physical-band theorem, and success in an averaged or smoothed model must not be reported as success at the fixed physical endpoint.

12 Exact certificate and claim ledger

The reproducibility directory contains

`experiments/tpc45_certificate.py.`

It uses the Python standard library, exact Gaussian-integer arithmetic, exhaustive finite sign cubes, and `fractions.Fraction`. A successful run writes `experiments/tpc45_certificate.json`. The checked families are:

- (i) the actual toy labels $u = d(\ell dj + 1)$, the packet-defined band, uniqueness of a band prime, exact physical sign deletion, and equality of the original and grouped energies on every remaining sign environment;
- (ii) the grouped diagonal inequality and equality in the abstract fiber-sum operator norm;
- (iii) the physical-low affine form on every high-prime environment;

- (iv) the low-averaged pure Gram projection and its all-minus synthesis identity on exhaustive low and high cubes;
- (v) the fixed-determinant ladder, its affine parameter step, and the common physical sign on an equal-residual fiber; and
- (vi) every rational exponent and endpoint margin used in the paper.

The current certificate performs 32894 exact checks. Its canonical payload hash is

e9c3bdf309863cf560fd31e5253e1429236281069101df26c7fe689d6faa438e.

The hash covers the report before the hash field itself is appended.

The certificate is deliberately finite. It checks identities and arithmetic ledgers, not the assertions that $\tau(n) = X^{o(1)}$, that primes exist in each fixed power subinterval, or that a density-one probability tends to one. Those are proved analytically in the text. Most importantly, it does not sample or certify Möbius cancellation, a fixed-determinant Bessel estimate, or a prime-pair asymptotic.

For clarity, the logical status of the main statements is as follows.

1. The actual coefficient expansion, prime-incidence alternatives, cofactor deletion, fiber bounds, grouped diagonal bounds, affine Gram forms, and determinant-ladder parameterization are unconditional finite or asymptotic theorems under the inherited TPC packet hypotheses.
2. The mesoscopic physical-band freezing and its two-parameter band/cutoff tradeoff are unconditional consequences of those theorems and the robust-face argument.
3. The larger cutoff $(\log X)^{1003/1000}$ applies to the shortened touched component. It is not claimed for the untouched part or the full packet.
4. The fixed-determinant synthesis estimate in [section 11](#) is a proposed next input, not a theorem of this paper.
5. The comparisons in [section 10](#) are applicability audits. A mismatch with a cited theorem is not an impossibility result for every future method.
6. No statement here implies the twin-prime conjecture, a Hardy–Littlewood asymptotic, a breach of the sieve parity barrier, or the complete deterministic TPC endpoint.

Thus the positive result is a real partial de-randomization of actual TPC coefficients, while the remaining gap is displayed rather than silently promoted to an assumption.

References

- [1] Sandro Bettin and Vorrapan Chandee. Trilinear forms with Kloosterman fractions. *Advances in Mathematics*, 328:1234–1262, 2018. doi: 10.1016/j.aim.2018.01.026. URL <https://doi.org/10.1016/j.aim.2018.01.026>.
- [2] Jean Bourgain, Peter Sarnak, and Tamar Ziegler. Disjointness of Möbius from horocycle flows. In Hershel M. Farkas, Marvin I. Knopp, Robert C. Gunning, and B. A. Taylor, editors, *From Fourier Analysis and Number Theory to Radon Transforms and Geometry*, volume 28 of *Developments in Mathematics*, pages 67–83. Springer, 2013. doi: 10.1007/978-1-4614-4075-8_5. URL https://doi.org/10.1007/978-1-4614-4075-8_5.
- [3] Anji Dong, Nicolas Robles, and Dirk Zeindler. Bilinear forms with Kloosterman fractions and applications. Withdrawn preprint, version 2; the withdrawal note states that a missing factor invalidates the claimed improved bound, 2026. URL <https://arxiv.org/abs/2601.00292>.

- [4] Étienne Fouvry and Maksym Radziwiłł. Level of distribution of unbalanced convolutions. *Annales Scientifiques de l'École Normale Supérieure*, 55(2):537–568, 2022. doi: 10.24033/asens.2501. URL <https://doi.org/10.24033/asens.2501>.
- [5] Liang Wang. Annealed multiplicative closure at the four-Möbius gate: Prime-sign orthogonality, product-fiber rigidity, and the deterministic Walsh corner. Preprint, repository commit 301f7fc, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/301f7fc/papers/tpc-43-annealed-multiplicative-closure.
- [6] Liang Wang. Robust small-prime freezing at the four-Möbius gate: Walsh-face compression, exact prime descent, and the high-prime drift barrier. Preprint, repository commit b4f1a69, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/b4f1a69/papers/tpc-44-robust-small-prime-freezing.
- [7] Thomas Wright. Trilinear Kloosterman fractions I: Partially fixed moduli and unbalanced convolutions. Preprint, 2026. URL <https://arxiv.org/abs/2604.25177>.